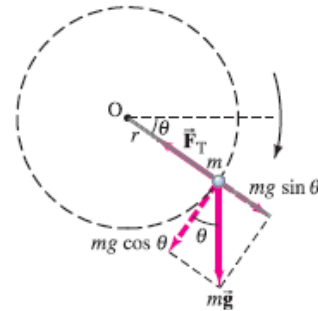


Tutorial 5: Uniform Circular Motion, Torque and Rotational Motion

Q1.

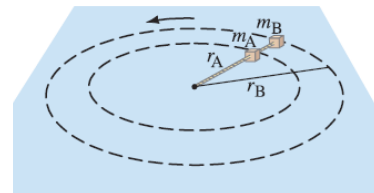
A bucket of mass $m = 2.00$ kg is whirled in a vertical circle of radius $r = 1.10$ m. At the lowest point of its motion the tension in the rope supporting the bucket is 25.0 N.

- Find the tangential speed of the bucket.
- What should the speed of the bucket be at the top of the circle so that the rope is not slack?
- What are the tangential and radial components of net force on the bucket, as functions of θ (see figure)?



Q2.

Two blocks of masses m_A and m_B , are connected to each other by cords as shown. They rotate about the central axis at frequency f on a frictionless surface at distances r_A and r_B from the centre. Derive an algebraic expression for the tension in each segment of the cord.



Q3. The blades of a blender rotate at a constant rate of 650 rpm.

The radius of each blade is 0.055 m.

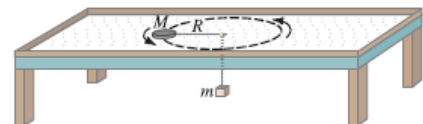
- What is the angular velocity in rad/s?
 - What is the linear speed and centripetal acceleration of a point on the end of a blade?
- When the motor is turned off, the blades slow to a rest in 4.0 s.
- What is the angular acceleration as the blades slow down?

Q4.

A coin placed 12.0 cm from the axis of a rotating turntable of variable speed. When the speed of the turntable is slowly increased, the coin remains fixed on the turntable until a rate of 35.0 rpm is reached, at which point the coin slides off. What is the coefficient of static friction between the coin and the turntable?

Q5.

A flat puck (mass M) is revolved in a circle on a frictionless air hockey table top, and is held in this orbit by a light cord which is connected to a dangling mass (mass m) through a central hole as shown. Show that the speed of the puck is given by $v = (mgR/M)^{1/2}$.



Q6.

A banked curve of radius R in a new highway is designed so that a car traveling at speed v_0 can negotiate the turn safely on ice (zero friction). If the car travels too slowly, it tends to slip towards the center of the circle. If it travels too fast, it tends to slip away from the center of the circle. If the coefficient of static friction increases, it becomes possible for a car to stay on the road while travelling within a range of speeds, from $v_{\min}$ to $v_{\max}$. Derive formulas for these speeds as a function of μ_s , v_0 and R .

Answers

- | | | | |
|-----|---|-----|--|
| Q1. | (a) 1.72 m/s , (b) 3.28 m/s | Q2. | $4\pi^2 m_B r_B f^2, 4\pi^2 f^2 (m_A r_A + m_B r_B)$ |
| Q3. | (a) 68.1 rad s ⁻¹ , (b) 3.74 m s ⁻¹ , 255 m s ⁻² (c) -17.0 rad s ⁻² | Q4. | 0.164 |
| Q6. | $v_0 \sqrt{\frac{(1 - \mu_s Rg/v_0^2)}{(1 + \mu_s v_0^2/Rg)}}, v_0 \sqrt{\frac{(1 + Rg\mu_s/v_0^2)}{(1 - \mu_s v_0^2/Rg)}}$ | | |

Note: There will be a midterm (10%) in class during tutorial.